



SYSTEM INTERFACES

STAKEHOLDER REFERENCE GUIDE

By SPP IT

Published on January 29, 2025

Revised on July 30, 2026

REVISION HISTORY

DATE	AUTHOR	CHANGE DESCRIPTION
1/29/2025	SPP IT	Initial Publication This reference guide replaces the following documents previously posted on SPP.org: • SPP External Technical Documents Inventory • SPP User Interface Guide Inventory • SPP Member Facing Technology Stack • WEIS Technical Data Exchange Preview
7/30/2026	SPP IT	Updated document to include the following: • New central location for accessing all User Guides, APIs, and Integration documentation • Markets+ acronym • Markets+ documentation currently available on SPP.org (additional documentation will be added throughout the Markets+ implementation) Removed decommissioned WEIS documentation

CONTENTS

REVISION HISTORY	I
APPLICABLE SERVICES	1
INTRODUCTION	1
Purpose	1
Intended Audience	2
Document Structure	2
Terms & Conditions	2
Acronyms	2
SPP SYSTEM INTERFACES	3
Technology Stack	4
User Interfaces	4
Programmatic Interfaces	5
System Updates & Maintenance	5
INTEGRATION AND USER GUIDES	5
General Technical Specifications	6
System-Specific API Specifications	8
Public Data Specifications	11
User Interface Guides	12
Data Exchange Validations	15
TECHNICAL SUPPORT	16
APPENDIX A: RTO DOCUMENTATION	17
APPENDIX B: ENGINEERING DOCUMENTATION	28
APPENDIX C: MARKETS+ DOCUMENTATION	31
APPENDIX D: WEST RC DOCUMENTATION	38
APPENDIX E: WEIS MARKET DOCUMENTATION	44

APPLICABLE SERVICES

The following SPP services are within the scope of the integrations and processes described in this document.

SERVICE	IN SCOPE
Engineering	☒
Markets+	☒
RTO (Integrated Marketplace / Transmission Services)	☒
WEIS Market	☒
Western Reliability Coordination	☒

INTRODUCTION

The term “stakeholder” refers to all stakeholders, which may include market participants, members, customers, transmission owners/operators, transmission customers, meter agents, generators, load serving entities, vendors, etc.

PURPOSE

This guide provides information on the technical documentation available for the stakeholder-facing interfaces managed by SPP that allow for the exchange of data between stakeholders and SPP systems.

There are graphical user interfaces (UIs) and programmatic interfaces for SPP’s stakeholder-facing systems and applications. In most cases, both a UI and programmatic interface are provided for each system and generally support the same functions. However, some exceptions do exist, as user interfaces and/or programmatic interfaces are not required in all cases. For example, the Schedule Data API provides only API access while the Ratings Submission Tool offers only UI access.

Any access or exchange of data outside of officially published SPP interface specifications is neither supported nor encouraged.

INTENDED AUDIENCE

This document is intended for end users of SPP’s accessible User Interfaces (UIs), as well as software developers, technical analysts, and integration specialists responsible for designing, implementing, and supporting data exchange interfaces with SPP.

DOCUMENT STRUCTURE

This document is structured into sections that assist stakeholders by presenting commonly used acronyms related to integrations, detailing the technology stack implemented by SPP, and providing a thorough inventory of integration resources available on SPP.org. These resources encompass guides for SPP’s user interfaces, programmatic interfaces, and data exchange validation rules. Additionally, a set of appendices located at the end of the document offers a comprehensive list of documents applicable to various SPP services, such as RTO and Markets+.

TERMS & CONDITIONS

SPP’s copyright, disclaimer, privacy policy, and acceptable use policy are available from the Terms & Conditions page on SPP.org.

[Home > Terms & Conditions](#)

ACRONYMS

ACRONYM	DEFINTION
API	Application Programmatic Interface
ENG	Engineering Services
FTP	File Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
M+	Markets+ Service
PMU	Phasor Measurement Unit

REST	Representational State Transfer
RTO	Regional Transmission Organization
SOAP	Simple Object Access Protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
UAA	User Authorization and Authentication
UI	Browser-based User Interface
WEIS	Western Energy Imbalance Market Services
WITT	WECC Interchange Tool
WRC	Western Reliability Coordination services

SPP SYSTEM INTERFACES

The following systems are available to stakeholders' exchanging information with SPP. For additional information, please refer to the documents described in the Integration and User Guides section of this document.

SYSTEM INTERFACE	INTERFACE TYPE	SERVICE
Credit & Risk		
Credit Management System (CMS)	UI, API	RTO, WEIS, M+
Credit Stacking		
Credit Stacking System (CSS) - Z2	UI	RTO
Engineering		
Engineering Data Submission Tool (EDST)	UI	RTO, ENG
Hosting Capacity Tool (HCT)	UI	RTO, ENG
SmartQ	UI	RTO, ENG
Transmission Reporting & Communications (TRAC) TAGIT/SCERT	UI	RTO, ENG
Market Monitoring		
Market Monitoring Portal (MMP)	UI	RTO, M+
Market		
Congestion Rent Tool (CRT)	UI, API	M+

RTO Day-Ahead and Real-Time Markets System	UI, API	RTO
M+ Day-Ahead and Real-Time Markets System	UI, API	M+
Transmission Congestion Rights Market (TCR)	UI, API	RTO
Outage Management		
Outage Management System (CROW)	UI, API	RTO, M+, WRC
Portal		
SPP Portal	UI	RTO, ENG, M+, WEIS, WRC
Public Reports		
Markets & Operations	UI, sFTP	RTO, M+, WEIS
Settlements	UI, sFTP	RTO, M+, WEIS
Real-Time Operations		
Automatic Generation Controls (AGC)	TCP/IP	RTO, M+, WRC
Dispatcher Training Simulator (DTS)	UI	RTO, M+, WRC
ICCP Submission Tool (ICCPST)	UI	RTO, M+, WRC
Inter-Control Center Communications Protocol (ICCP)	TCP/IP	RTO, M+, WRC
Limit Exchange Portal (LEP)	UI, API	RTO, M+, WRC
Phasor Measurement Unit (PMU)	TCP/IP	RTO, WRC
Ratings Submission Tool (RST)	UI	RTO, M+, WRC
Replicated Data Server (RDS)	UI	RTO, M+, WRC
Ventura WIT Interchange Checkout (EIDE Data)	API	RTO, M+, WRC
West Load/Gen Forecast (PutSchedule)	API	RTO, WRC
Settlements		
Market Transmission Usage (MTU)	TBD	M+
Settlement Management System (SMS)	UI, API	RTO, M+, WEIS
Schedule Data	API	RTO
Training & Support		
Learning Management System (LMS)	UI	RTO, ENG, M+, WEIS, WRC
Request Management System (RMS)	UI	RTO, ENG, M+, WEIS, WRC
User Authentication & Authorization		
User Authentication & Authorization (UAA)	UI, API	RTO, ENG, M+, WEIS, WRC

TECHNOLOGY STACK

USER INTERFACES

SPP User Interfaces (UIs) are compatible with the latest versions of Chrome and Firefox. Use of any other browser is not recommended or supported.

PROGRAMMATIC INTERFACES

SPP utilizes SOAP and REST-based web services over HTTPS for programmatic interactions. Most of these services require two-factor authentication. SPP also supports the use of TCP/IP for interfacing with AGC, ICCP, and PMU as well as FTP for the programmatic retrieval of Public Data.

SYSTEM UPDATES & MAINTENANCE

All hardware and software updates as well as maintenance performed on stakeholder-facing SPP systems are published to the [Change Calendar](#) on the SPP Portal.

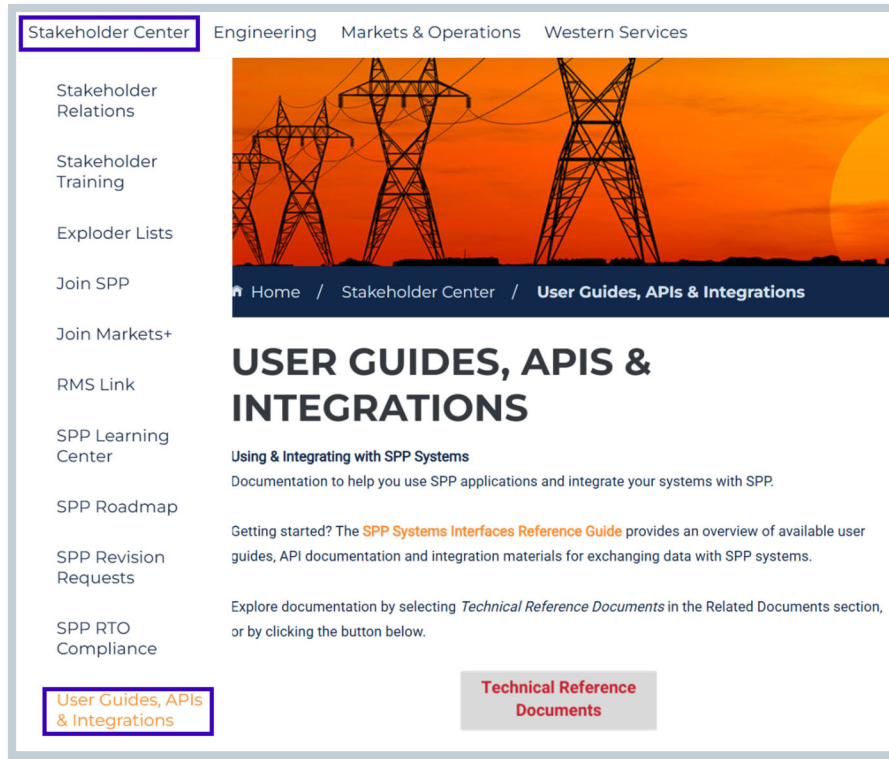
Any change that requires stakeholders to implement system and/or process changes is also reviewed in the appropriate user forum:

- Change User Forum – for changes impacting SPP’s RTO, Engineering, and West RC systems
- Markets+ Change User Forum – for changes impacting Markets+ systems

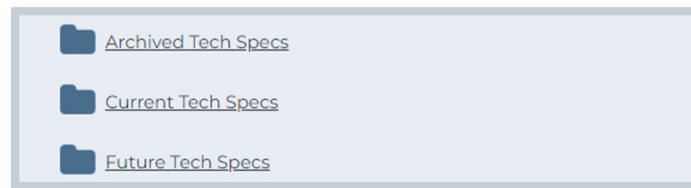
INTEGRATION AND USER GUIDES

This section provides information on the technical documentation and user guides available to stakeholders. Unless otherwise specified, all system documents are housed within the Technical Reference Documents section on SPP.org.

Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents



To facilitate the publication of incremental versions of SPP system documentation, the following folder structure is typically utilized.



The **Current** folder contains documentation for current production versions of system interfaces. The **Future** folder includes documentation for soon-to-be-production versions, and the **Archived** folder holds retired versions of the documents.

GENERAL TECHNICAL SPECIFICATIONS

The following documentation is applicable to multiple SPP systems.

Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents

DOCUMENT	DESCRIPTION	SERVICE
Marketplace External Stakeholder Data Points	Provides an inventory of data points per RTO system and information on the methods used for exchange of the data.	RTO
Portal Application Role Reference Guide	Guide to assist Local Security Administrators (LSA) with setting up users in the User Authorization and Authentication. Note: This guide is available to LSAs at the following location in the SPP Portal. <i>Portal > Administration > User Administration > LSA Documentation</i>	RTO, ENG, WRC, WEIS, M+
SPP Connectivity Specifications	Describes the connectivity requirements and details such as certificate information and URLs for SPP environments. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs folder.</i>	RTO, ENG, WRC, WEIS, M+
SPP UAA API Two-Factor Authentication Technical Specifications	Describes the authentication factors required for accessing SPP systems. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs folder.</i>	RTO, ENG, WRC, WEIS, M+

SPP UAA API Specifications	Defines SPP's user authorization and authentication API specifications. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs folder.</i>	RTO, ENG, WRC, WEIS, M+
WEIS External Stakeholder Data Points	Provides an inventory of data points per system and information on the methods used for exchange of the data.	WEIS

SYSTEM-SPECIFIC API SPECIFICATIONS

The following documentation is posted on SPP.org in the **Tech Specs** directory unless otherwise noted.

Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

SYSTEM	DOCUMENT	DESCRIPTION	SERVICE
AGC	AGC Integration Guide	Technical specifications for the Automatic Generation Controls. This document is password protected. Please submit a ticket via RMS to request the password. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > AGC Integration Guide</i>	RTO, WRC
CMS	SPP Credit Limit Report API Specifications	Credit Limit Report API Data Exchange Guide and schemas for Market+ Credit.	M+

CMS	Credit Limit Report API Specifications	Technical specifications for the Credit Limit Report API.	RTO
CROW	SPP CROW API Specifications	Technical specifications for the CROW API.	RTO, M+, WRC
CROW	CROW Access Request Instructions and CROW Access Request Form	Instructions and form for requesting access to CROW. Note: This documentation is available at the following location on SPP.org. <i>SPP Documents > Operations > Outage Coordination</i>	RTO, M+, WRC
CRT	Markets Plus CRT API Specifications	Congestion Rent Tool API Data Exchange Guide and schemas for Market+ Congestion Rent.	M+
CSS	CSS Specifications	Report definitions, schema definitions and sample files for the Credit Stacking System (Z2) processing.	RTO
ICCP	ICCP Handbook	Technical ICCP documentation for data exchanged between power system control centers. This document is password protected. Please submit a ticket via RMS to request the password. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > ICCP Handbook</i>	RTO, WRC, M+
LEP	SPP LEP API Data Exchange Guide	Technical specifications for the LEP API	RTO, WRC, M+

Markets	RTO Markets API Specifications	Data Exchange Guide and associated WSDLs, XSDs, and Specifications for SPP Integrated Marketplace API web and notification services.	RTO
Markets	Markets Plus Markets API Specifications	Markets System API Data Exchange Guide and schemas for Market+ Markets.	M+
PMU	SPP PMU Communication Handbook	Defines the practices, conventions, and fundamental parameters required for PMU data to be exchanged with SPP. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > PMU Handbook</i>	RTO, WRC
Schedule Data	Schedule Data API Specifications	Technical specifications for the Schedule Data API.	RTO
SMS	Settlement Management System API XML Specifications	Settlement System Web Services Data Exchange Guide, WSDLs, XSDs, and sample files for Market and Transmission Settlements.	RTO
SMS	Markets Plus Settlements API Specifications	Settlement System API Data Exchange Guide, schemas, and sample files for Market+ Settlements.	M+
SMS	WEIS Settlement Management System API XML Specifications	Settlement System Web Services Data Exchange Guide, WSDLs, XSDs, and sample files for WEIS Market Settlements.	WEIS
TCR Market	TCR API Specifications	Technical specifications for the TCR Market APIs.	RTO

West Load/Gen Forecast (PutSchedule)	SPP PutSchedule API Specifications	Technical specifications for the SPP EIDE Put Schedule API.	RTO, WRC
WIT Interchange Checkout	SPP EIDE Put Meter Service API Specifications	Technical specifications for the SPP EIDE Put Meter API. EIDE Details can be found at: https://devws.wit.oati.com/EIDEWS.aspx	RTO, WRC

PUBLIC DATA SPECIFICATIONS

The following documentation is posted on SPP.org in the **Public Data** subfolder.

Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data

SYSTEM	DOCUMENT	DESCRIPTION	SERVICE
All	SPP Public Data Access	Reference guide for accessing public data via the SPP Portal and FTP.	RTO, WEIS, M+
Markets, Operations, TCR Market	SPP Markets Public Data Guide and Samples	Reference guide and sample files for SPP RTO Markets and Operations Public Data offerings, including Real-Time Balancing Market, Day-Ahead Market, TCR Market, and operational data.	RTO
Market, Operations	SPP WEIS Public Data Guide	Reference guide and sample files for SPP WEIS Market and Operations Public Data offerings, including Real-Time Balancing Market and operational data.	WEIS

SMS	Settlements Public Data Guide	Reference guide for Settlements Public Data offerings such as Settlement calendars and rates.	RTO
SMS	WEIS Settlement Public Data Guide	Reference guide for WEIS Settlement Public Data offerings such as Settlement calendars and rates.	WEIS
Transoutage	Transoutage Specifications	This guide describes the download functionality and technical details to allow users to automate pulling this data.	RTO

USER INTERFACE GUIDES

The following documentation is posted on SPP.org in the **User Interface Guides** subfolder unless otherwise noted.

Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides

SYSTEM	DOCUMENT	DESCRIPTION	SERVICE
CROW	SPP CROW Web GUI Tutorial	Comprehensive tutorial explaining the procedure for entry of outages and managing existing outages in CROW. Note: This guide is available at the following location on SPP.org. <i>SPP Documents > Operations > Outage Coordination</i>	RTO, M+, WRC
CRT	Markets Plus CRT User Interface Guide	This guide describes how to submit, and if applicable, approve input data, register for CRT notifications, view reports, and download data.	M+

CSS	CSS User Interfaces Guide	Reference guide for Transmission Customers and Transmission Owners that describes how to view and download CSS Input Data and CSS Output Reports from the SPP Portal.	RTO
DTS	SPP DTS User Guide	Reference guide is not publicly available. Please submit a ticket via RMS to request additional information for this application.	RTO, WRC
EDST	EDST User Guide	Reference guide describing how to use the Engineering Data Submission Tool (EDST) for managing data used for the transmission planning models, NERC reporting requirements, and SPP Tariff reporting requirements. Note: This guide is available at the following locations on SPP.org. Model Development - <i>SPP Documents > Stakeholder Group Documents > Model Development Advisory Group > MDAG Reference Documents > EDST MD Materials</i> Resource Adequacy users - <i>SPP Documents > Engineering & Planning > Resource Adequacy > EDST Materials</i>	RTO, ENG
HCT	Hosting Capacity Tool UI Guide and FAQ	Reference guide for users utilizing the Hosting Capacity Tool (HCT), which is a Pre-screening Tool (PST) for external users. This guide describes how to create a new user account and generate a map color contour from user input.	RTO, ENG
ICCPST	ICCP Online Submission Tool User Manual External	Reference guide for users of the ICCP Submission Tool.	RTO, WRC, M+

LEP	SPP LEP User Interface Guide	Describes interaction with the SPP implementation of GEV's LEP (General Electric Vernova's Limit Exchange Portal). This document walks users through the UI and its various functions.	RTO, WRC, M+
Markets	RTO Markets User Interface Guide	Reference guide for Market Participants that describes how to submit bids and offers, as well as view Market results, for the SPP Integrated Marketplace using the MUI in the SPP Portal.	RTO
RDS	SPP RDS User Guide	Reference guide is not publicly available. Please submit a ticket via RMS to request additional information for this application.	RTO, WRC
RMS	RMS Job Aid	Reference guide that describes how to access and submit inquiries to SPP via RMS. Note: This guide is available at the following location on SPP.org. <i>Stakeholder Center > Stakeholder Relations > Request Management System</i>	RTO, ENG, WRC, WEIS, M+
RST	SPP Ratings Submission Tool User Interface Guide	Reference guide that describes how to access and submit line rating changes to SPP.	RTO, WRC
SmartQ	SmartQ DPA Tutorial	Reference guide that describes how to access and submit Delivery Point Addition (DPA) requests to SPP. Note: This guide is available at the following location on SPP.org. <i>SPP Documents > Engineering & Planning > Tariff Studies > Delivery Point Addition Studies</i>	RTO, ENG

SMS	Settlement User Interfaces Guide	This guide describes how to submit, and if applicable, approve input data, register for Settlement notifications, view meter reports, and Settlement results (e.g. invoices, statements) via UIs available in the SPP Portal.	RTO
SMS	WEIS Settlement User Interfaces Guide	This guide describes how to submit, register for event notifications, view meter reports, and WEIS Market Settlement results (e.g. invoices, statements) via UIs available in the SPP Portal.	WEIS
TCR Market	TCR MUI User Manual	Provides Market Participants and Asset Owners with an introduction to SPP's TCR Market system and its user interfaces available in the SPP Portal.	RTO
TRAC/ TAGIT/SCERT	TRAC User Guide TRAC Data Dictionary	Reference guide describing how to use the Transmission Reporting & Communications (TRAC) Tool to facilitate project tracking, the SPP Transmission Expansion Plan (STEP), and issuance of notification to construct (NTC). Note: This guide is available at the following location on SPP.org. <i>SPP Documents > Stakeholder Group Documents > Project Cost Working Group > PCWG Resource Documents</i>	RTO, ENG

DATA EXCHANGE VALIDATIONS

Many SPP systems validate stakeholder data submissions upon receipt. Any data that does not adhere to the defined business rules is rejected and a validation error returned to the submitter. The data validation rules are available on SPP.org in the **Business Rules Guides** folder.

Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides

SYSTEM	DOCUMENT	DESCRIPTION	SERVICE
LEP	SPP Limit Exchange Portal (LEP) Data Exchange Validations	Defines the business rules and validations applied to the various submissions available to LEP if the logged in user is the ratings provider for selected facilities for network segments and has “write” access.	RTO, WRC, M+
Markets	Integrated Marketplace Market User Interface Business Validations	Defines the business validation rules associated with participant data submissions to the SPP Integrated Marketplace.	RTO
Markets	Integrated Marketplace Resource Offer Roll Forward Guide	Describes how the submission of resource offer data will be carried forward and applied to future operating days in the SPP Integrated Marketplace.	RTO
SMS	Settlement Member Data Exchange Validations	Defines the business validation rules associated with stakeholder data submissions to the SPP Markets and Transmission Settlement system.	RTO
SMS	WEIS Settlement Member Data Exchange Validations	Defines the business validation rules associated with stakeholder data submissions to the SPP WEIS Market Settlement system.	WEIS

TECHNICAL SUPPORT

To request additional information or receive technical support, please submit a ticket via the SPP Request Management System (RMS). Additional information on the SPP RMS may be found on the Stakeholder Relations page on SPP.org.

Stakeholder Center > Stakeholder Relations

APPENDIX A: RTO DOCUMENTATION

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Data Points	Marketplace External Stakeholder Data Points Provides an inventory of data points per RTO system and information on the methods used for exchange of the data. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents
General	Connectivity	SPP Connectivity Specifications Describes the connectivity requirements and details such as certificate information and URLs for SPP environments. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Authentication / Authorization	Portal Application Role Reference Guide Guide to assist Local Security Administrators (LSA) with setting up users in the User Authorization and Authentication. Note: This guide is available to LSAs at the following location in the SPP Portal. Portal > Administration > User Administration > LSA Documentation

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Authentication / Authorization	SPP UAA API Two-Factor Authentication Technical Specifications Describes the authentication factors required for accessing SPP systems. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Authentication / Authorization	SPP UAA API Specifications Defines SPP's user authorization and authentication API specifications. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
API Specs	AGC	AGC Integration Guide Technical specifications for the Automatic Generation Controls. This document is password protected. Please submit a ticket via RMS to request the password. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > AGC Integration Guide
API Specs	CROW	SPP CROW API Specifications Technical specifications for the CROW API. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
API Specs	CROW	CROW Access Request Instructions and CROW Access Request Form SPP Documents > Operations > Outage Coordination
API Specs	CSS	CSS Specifications Report definitions, schema definitions and sample files for the Credit Stacking System (Z2) processing. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	ICCP	ICCP Handbook Technical ICCP documentation for data exchanged between power system control centers. This document is password protected. Please submit a ticket via RMS to request the password. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > ICCP Handbook
API Specs	LEP	SPP LEP API Data Exchange Guide Technical specifications for the LEP API Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	Markets	RTO Markets API Specifications Data Exchange Guide and associated WSDLs, XSDs, and Specifications for SPP

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Integrated Marketplace API web and notification services. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	PMU	SPP PMU Communication Handbook Defines the practices, conventions, and fundamental parameters required for PMU data to be exchanged with SPP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > PMU Handbook
API Specs	Schedule Data	Schedule Data API Specifications Technical specifications for the Schedule Data API. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	SMS	Settlement Management System API XML Specifications Settlement System Web Services Data Exchange Guide, WSDLs, XSDs, and sample files for Market and Transmission Settlements. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	TCR Market	TCR API Specifications Technical specifications for the TCR Market APIs.

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	West Load/Gen Forecast (PutSchedule)	SPP PutSchedule API Specifications Technical specifications for the SPP EIDE Put Schedule API. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	WIT Interchange Checkout	SPP EIDE Put Meter Service API Specifications Technical specifications for the SPP EIDE Put Meter API. https://devws.wit.oati.com/EIDEWS.asmx
Public Data	All	SPP Public Data Access Reference guide for accessing public data via the SPP Portal and FTP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
Public Data	Markets, Operations, TCR Market	SPP Markets Public Data Guide and Samples Reference guide and sample files for SPP RTO Markets and Operations Public Data offerings, including Real-Time Balancing Market, Day-Ahead Market, TCR Market, and operational data.

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
Public Data	SMS	Settlements Public Data Guide Reference guide for Settlements Public Data offerings such as Settlement calendars and rates. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
Public Data	Transoutage	Transoutage Specifications This guide describes the download functionality and technical details to allow users to automate pulling this data. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
UI Guides	CROW	SPP CROW Web GUI Tutorial Comprehensive tutorial explaining the procedure for entry of outages and managing existing outages in CROW. SPP Documents > Operations > Outage Coordination
UI Guides	CSS	CSS User Interfaces Guide Reference guide for Transmission Customers and Transmission Owners that describes how to view and download CSS Input Data and CSS Output Reports from the SPP Portal.

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	DTS	SPP DTS User Guide Reference guide is not publicly available. Please submit a ticket via RMS to request additional information for this application.
UI Guides	EDST	EDST User Guide Reference guide describing how to use the Engineering Data Submission Tool (EDST) for managing data used for the transmission planning models, NERC reporting requirements, and SPP Tariff reporting requirements. Model Development - SPP Documents > Stakeholder Group Documents > Model Development Advisory Group > MDAG Reference Documents > EDST MD Materials Resource Adequacy users - SPP Documents > Engineering & Planning > Resource Adequacy > EDST Materials
UI Guides	HCT	Hosting Capacity Tool UI Guide and FAQ Reference guide for users utilizing the Hosting Capacity Tool (HCT), which is a Pre-screening Tool (PST) for external users. This guide describes how to create a new user account and generate a map color contour from user input. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
UI Guides	ICCPST	ICCP Online Submission Tool User Manual External Reference guide for users of the ICCP Submission Tool. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	LEP	SPP LEP User Interface Guide Describes interaction with the SPP implementation of GEV's LEP (General Electric Vernova's Limit Exchange Portal). This document walks users through the UI and its various functions. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	Markets	RTO Market User Interface Guide Reference guide for Market Participants that describes how to submit bids and offers, as well as view Market results, for the SPP Integrated Marketplace using the MUI in the SPP Portal. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	RDS	SPP RDS User Guide Reference guide is not publicly available. Please submit a ticket via RMS to request additional information for this application.

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	RMS	RMS Job Aid Reference guide that describes how to access and submit inquiries to SPP via RMS. Stakeholder Center > Stakeholder Relations > Request Management System
UI Guides	RST	Ratings Submission Tool User Interface Guide Reference guide that describes how to access and submit line rating changes to SPP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	SmartQ	SmartQ DPA Tutorial Reference guide that describes how to access and submit Delivery Point Addition (DPA) requests to SPP. SPP Documents > Engineering & Planning > Tariff Studies > Delivery Point Addition Studies
UI Guides	SMS	Settlement User Interfaces Guide This guide describes how to submit, and if applicable, approve input data, register for Settlement notifications, view meter reports, and Settlement results (e.g. invoices, statements) via UIs available in the SPP Portal.

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	TCR Market	SPP TCR MUI User Manual Provides Market Participants and Asset Owners with an introduction to SPP's TCR Market system and its user interfaces available in the SPP Portal. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	TRAC/ TAGIT/SCERT	TRAC User Guide / TRAC Data Dictionary Reference guide describing how to use the Transmission Reporting & Communications (TRAC) Tool to facilitate project tracking, the SPP Transmission Expansion Plan (STEP), and issuance of notification to construct (NTC). SPP Documents > Stakeholder Group Documents > Project Cost Working Group > PCWG Resource Documents
Data Exchange Validations	LEP	SPP Limit Exchange Portal (LEP) Data Exchange Validations Defines the business rules and validations applied to the various submissions available to LEP if the logged in user is the ratings provider for selected facilities for network segments and has "write" access. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides
Data Exchange Validations	Markets	Integrated Marketplace Market User Interface Business Validations

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
		Defines the business validation rules associated with participant data submissions to the SPP Integrated Marketplace. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides
Data Exchange Validations	Markets	Integrated Marketplace Resource Offer Roll Forward Guide Describes how the submission of resource offer data will be carried forward and applied to future operating days in the SPP Integrated Marketplace. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides
Data Exchange Validations	SMS	Settlement Member Data Exchange Validations Defines the business validation rules associated with stakeholder data submissions to the SPP Markets and Transmission Settlement system. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides

APPENDIX B: ENGINEERING DOCUMENTATION

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Connectivity	SPP Connectivity Specifications Describes the connectivity requirements and details such as certificate information and URLs for SPP environments. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Authentication / Authorization	Portal Application Role Reference Guide Guide to assist Local Security Administrators (LSA) with setting up users in the User Authorization and Authentication. Note: This guide is available to LSAs at the following location in the SPP Portal. Portal > Administration > User Administration > LSA Documentation
General	Authentication / Authorization	SPP UAA API Two-Factor Authentication Technical Specifications Describes the authentication factors required for accessing SPP systems. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs

General	Authentication / Authorization	SPP UAA API Specifications Defines SPP’s user authorization and authentication API specifications. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
UI Guides	EDST	EDST User Guide Reference guide describing how to use the Engineering Data Submission Tool (EDST) for managing data used for the transmission planning models, NERC reporting requirements, and SPP Tariff reporting requirements. Model Development - SPP Documents > Stakeholder Group Documents > Model Development Advisory Group > MDAG Reference Documents > EDST MD Materials Resource Adequacy users - SPP Documents > Engineering & Planning > Resource Adequacy > EDST Materials
UI Guides	HCT	Hosting Capacity Tool UI Guide and FAQ Reference guide for users utilizing the Hosting Capacity Tool (HCT), which is a Pre-screening Tool (PST) for external users. This guide describes how to create a new user account and generate a map color contour from user input. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	RMS	RMS Job Aid Reference guide that describes how to access and submit inquiries to SPP via RMS.

		Stakeholder Center > Stakeholder Relations > Request Management System
UI Guides	SmartQ	SmartQ DPA Tutorial Reference guide that describes how to access and submit Delivery Point Addition (DPA) requests to SPP. SPP Documents > Engineering & Planning > Tariff Studies > Delivery Point Addition Studies
UI Guides	TRAC/ TAGIT/SCERT	TRAC User Guide / TRAC Data Dictionary Reference guide describing how to use the Transmission Reporting & Communications (TRAC) Tool to facilitate project tracking, the SPP Transmission Expansion Plan (STEP), and issuance of notification to construct (NTC). SPP Documents > Stakeholder Group Documents > Project Cost Working Group > PCWG Resource Documents

APPENDIX C: MARKETS+ DOCUMENTATION

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Connectivity	SPP Connectivity Specifications Describes the connectivity requirements and details such as certificate information and URLs for SPP environments. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Authentication / Authorization	Portal Application Role Reference Guide Guide to assist Local Security Administrators (LSA) with setting up users in the User Authorization and Authentication. Note: This guide is available to LSAs at the following location in the SPP Portal. Portal > Administration > User Administration > LSA Documentation
General	Authentication / Authorization	SPP UAA API Two-Factor Authentication Technical Specifications Describes the authentication factors required for accessing SPP systems. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Authentication / Authorization	SPP UAA API Specifications Defines SPP's user authorization and authentication API specifications. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
API Specs	CMS	SPP Credit Limit Report API Specifications Credit Limit Report API Data Exchange Guide and schemas for Market+ Credit. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	CROW	SPP CROW API Specifications Technical specifications for the CROW API. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
API Specs	CROW	CROW Access Request Instructions and CROW Access Request Form SPP Documents > Operations > Outage Coordination
API Specs	CRT	Markets Plus CRT API Specifications Congestion Rent Tool API Data Exchange Guide and schemas for Market+ Congestion Rent. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	ICCP	ICCP Handbook Technical ICCP documentation for data exchanged between power system control centers. This document is password protected. Please submit a ticket via RMS to request the password. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > ICCP Handbook

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
API Specs	LEP	SPP LEP API Data Exchange Guide Technical specifications for the LEP API Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	Markets	Markets Plus Markets API Specifications Markets System API Data Exchange Guide and schemas for Market+ Markets. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	SMS	Markets Plus Settlements API Specifications Settlement System API Data Exchange Guide, schemas, and sample files for Market+ Settlements. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
Public Data	All	SPP Public Data Access Reference guide for accessing public data via the SPP Portal and FTP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
UI Guides	CROW	SPP CROW Web GUI Tutorial Comprehensive tutorial explaining the procedure for entry of outages and managing existing outages in CROW. SPP Documents > Operations > Outage Coordination
UI Guides	CRT	Markets Plus CRT User Interface Guide This guide describes how to submit, and if applicable, approve input data, register for CRT notifications, view reports, and download data. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
UI Guides	ICCPST	ICCP Online Submission Tool User Manual External Reference guide for users of the ICCP Submission Tool. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	LEP	SPP LEP User Interface Guide Describes interaction with the SPP implementation of GEV's LEP (General Electric Vernova's Limit Exchange Portal). This document walks users through the UI and its various functions. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	RMS	RMS Job Aid Reference guide that describes how to access and submit inquiries to SPP via RMS. Stakeholder Center > Stakeholder Relations > Request Management System

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
Data Exchange Validations	LEP	SPP Limit Exchange Portal (LEP) Data Exchange Validations Defines the business rules and validations applied to the various submissions available to LEP if the logged in user is the ratings provider for selected facilities for network segments and has “write” access. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides

APPENDIX D: WEST RC DOCUMENTATION

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Connectivity	SPP Connectivity Specifications Describes the connectivity requirements and details such as certificate information and URLs for SPP environments. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Authentication / Authorization	Portal Application Role Reference Guide Guide to assist Local Security Administrators (LSA) with setting up users in the User Authorization and Authentication. Note: This guide is available to LSAs at the following location in the SPP Portal. Portal > Administration > User Administration > LSA Documentation
General	Authentication / Authorization	SPP UAA API Two-Factor Authentication Technical Specifications Describes the authentication factors required for accessing SPP systems. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Authentication / Authorization	SPP UAA API Specifications Defines SPP's user authorization and authentication API specifications. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
API Specs	AGC	AGC Integration Guide Technical specifications for the Automatic Generation Controls. This document is password protected. Please submit a ticket via RMS to request the password. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > AGC Integration Guide
API Specs	CROW	SPP CROW API Specifications Technical specifications for the CROW API. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
API Specs	CROW	CROW Access Request Instructions and CROW Access Request Form SPP Documents > Operations > Outage Coordination
API Specs	ICCP	ICCP Handbook Technical ICCP documentation for data exchanged between power system control centers. This document is password protected. Please submit a ticket via RMS to request the password. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > ICCP Handbook
API Specs	LEP	SPP LEP API Data Exchange Guide Technical specifications for the LEP API Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
API Specs	PMU	SPP PMU Communication Handbook Defines the practices, conventions, and fundamental parameters required for PMU data to be exchanged with SPP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > PMU Handbook
API Specs	West Load/Gen Forecast (PutSchedule)	SPP PutSchedule API Specifications Technical specifications for the SPP EIDE Put Schedule API. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs
API Specs	WIT Interchange Checkout	SPP EIDE Put Meter Service API Specifications Technical specifications for the SPP EIDE Put Meter API. https://devws.wit.oati.com/EIDEWS.aspx
UI Guides	CROW	SPP CROW Web GUI Tutorial Comprehensive tutorial explaining the procedure for entry of outages and managing existing outages in CROW. SPP Documents > Operations > Outage Coordination

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
UI Guides	DTS	SPP DTS User Guide Reference guide is not publicly available. Please submit a ticket via RMS to request additional information for this application.
UI Guides	ICCPST	ICCP Online Submission Tool User Manual External Reference guide for users of the ICCP Submission Tool. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	LEP	SPP LEP User Interface Guide Describes interaction with the SPP implementation of GEV's LEP (General Electric Vernova's Limit Exchange Portal). This document walks users through the UI and its various functions. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
UI Guides	RDS	SPP RDS User Guide Reference guide is not publicly available. Please submit a ticket via RMS to request additional information for this application. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
UI Guides	RMS	RMS Job Aid Reference guide that describes how to access and submit inquiries to SPP via RMS. Stakeholder Center > Stakeholder Relations > Request Management System
UI Guides	RST	Ratings Submission Tool User Interface Guide Reference guide that describes how to access and submit line rating changes to SPP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
Data Exchange Validations	LEP	SPP Limit Exchange Portal (LEP) Data Exchange Validations Defines the business rules and validations applied to the various submissions available to LEP if the logged in user is the ratings provider for selected facilities for network segments and has “write” access. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides

APPENDIX E: WEIS MARKET DOCUMENTATION

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Connectivity	SPP Connectivity Specifications Describes the connectivity requirements and details such as certificate information and URLs for SPP environments. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Authentication / Authorization	Portal Application Role Reference Guide Guide to assist Local Security Administrators (LSA) with setting up users in the User Authorization and Authentication. Note: This guide is available to LSAs at the following location in the SPP Portal. Portal > Administration > User Administration > LSA Documentation
General	Authentication / Authorization	SPP UAA API Two-Factor Authentication Technical Specifications Describes the authentication factors required for accessing SPP systems. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
General	Authentication / Authorization	SPP UAA API Specifications Defines SPP's user authorization and authentication API specifications. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Technical Specifications > Tech Specs
General	Data Points	WEIS External Stakeholder Data Points Provides an inventory of data points per system and information on the methods used for exchange of the data. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents
API Specs	SMS	WEIS Settlement Management System API XML Specifications Settlement System Web Services Data Exchange Guide, WSDLs, XSDs, and sample files for WEIS Market Settlements. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Tech Specs

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
Public Data	All	SPP Public Data Access Reference guide for accessing public data via the SPP Portal and FTP. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
Public Data	Market, Operations	SPP WEIS Public Data Guide Reference guide and sample files for SPP WEIS Market and Operations Public Data offerings, including Real-Time Balancing Market and operational data. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data
Public Data	SMS	WEIS Settlement Public Data Guide Reference guide for WEIS Settlement Public Data offerings such as Settlement calendars and rates. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Public Data

DOCUMENT CATEGORY	SYSTEM	DOCUMENT & LOCATION
UI Guides	RMS	RMS Job Aid Reference guide that describes how to access and submit inquiries to SPP via RMS. Stakeholder Center > Stakeholder Relations > Request Management System
UI Guides	SMS	WEIS Settlement User Interfaces Guide This guide describes how to submit, register for event notifications, view meter reports, and WEIS Market Settlement results (e.g. invoices, statements) via UIs available in the SPP Portal. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > User Interface Guides
Data Exchange Validations	SMS	WEIS Settlement Member Data Exchange Validations Defines the business validation rules associated with stakeholder data submissions to the SPP WEIS Market Settlement system. Stakeholder Center > User Guides, APIs & Integrations > Technical Reference Documents > Business Rules Guides